BEU CONNECT

Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2015

Time: 03 Hours

Code: 031609 ***Subject: POWER ELECTRONICS*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option (any seven): **[14]**

(a) When an SCR is reverse biased
- (i) two junctions are reverse biased and one junction is forward biased
- (ii) all the three junctions are reverse biased
- (iii) one junction is reverse biased and two junctions are forward biased
- (iv) Any of the above depending on the magnitude of reverse bias

(b) An R-C snubber circuit is used to protect a thyristor against
- (i) false triggering
- (ii) failure to turn on
- (iii) switching transients
- (iv) failure to commutate

(c) A single-phase semiconverter is feeding a highly inductive load and has free-wheeling diode across the load. The waveshapes of output voltage and output current
- (i) are similar
- (ii) are not similar
- (iii) may be similar or dissimilar
- (iv) are similar only if firing angle is zero

(d) A single-phase half-wave rectifier is feeding a resistive load. Input voltage $v = V_m \sin\omega t$. The output d.c. voltage is V_{dc} and output r.m.s. voltage is $V_{r.m.s.}$. If firing angle is 180°, V_{dc} and $V_{r.m.s.}$ respectively are
- (i) 0 and 0
- (ii) V_m/π and $V_m/\sqrt{(2)}$
- (iii) 0 and $-V_m$
- (iv) $-V_m/\pi$ and $-V_m/2$

(e) In a step-down chopper using pulse-width modulation $T_{on} = 3\times10^{-3}$s and $T_{off} = 1\times10^{-3}$s. The chopping frequency is
- (i) 333.33
- (ii) 500
- (iii) 250
- (iv) 1000

(f) In a three-phase full-wave regulator feeding a star-connected resistance load the input voltage is 400 V line to line. The firing angle is 160°. The line to line output voltage would be
- (i) 400 V
- (ii) about 20 V
- (iii) about 100 V
- (iv) zero

(g) Assertion (A): An inverter does not require forced commutation. Reason (R): A series inverter is a forced commutation inverter.
- (i) Both A and R are correct and R is correct explanation of A
- (ii) Both A and R are correct but R is not correct explanation of A
- (iii) A is correct but R is wrong
- (iv) A is wrong but R is correct

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(h) A three-phase bridge inverter is fed by 400 V battery. The load is star-connected and has a resistance of 10~Ω per phase. The peak value of load current is

(i) 8 A (iii) 10 A

(ii) 20 A (iv) 5 A

(i) The waveshape of output voltage of half-bridge inverter is

(i) sinusoidal (iii) triangular

(ii) square (iv) Either (i) or (ii)

(j) In dielectric heating, the thyristor circuits consist of

(i) rectifier-chopper combination (iii) controlled rectifier

(ii) a.c. regulator (iv) rectifier-inverter combination

Q.2 Solve both questions : **[6+8=14]**

(a) Discuss some of the advantages and disadvantages of power electronic converters.

(b) Define latching and holding currents as applicable to an SCR. Show these on the V-I characteristics.

Q.3 Solve both questions : **[6+8=14]**

(a) Why are dv/dt and di/dt protections in case of thyristor important?

(b) An SCR operating from a peak supply voltage of 400 V has the following specifications: Repetitive peak current I_p= 200A, $(di/dt)_{max}$= 50A/μs, $(dv/dt)_{max}$= 200V/μs. Choosing a factor of safety of 2 for I_p, $(di/dt)_{max}$ and $(dv/dt)_{max}$, design a suitable snubber circuit. The minimum value of load resistance is 10 ohms.

Q.4 Solve both questions : **[7+7=14]**

(a) Explain the principle of phase control with suitable diagram and waveform.

(b) Discuss in detail three-phase full converter with R-L-E load with suitable diagram and waveform.

Q.5 Solve both questions : **[7+7=14]**

(a) What is inverter? List a few industrial applications of inverter.

(b) Discuss in detail three-phase 120 degree mode voltage source inverter.

Q.6 Solve both questions : **[6+8=14]**

(a) Describe the principle of d.c. chopper operation.

(b) Derive an expression for its average d.c. output voltage.

Q.7 Solve both questions : **[8+6=14]**

(a) Describe integral cycle control type a.c. voltage controller.

(b) Enumerate the merits and demerits of a.c. voltage controller.

Q.8 Solve both questions : **[6+8=14]**

(a) Draw the basic integrated structure and the V-I characteristics of a TRIAC and briefly explain its principle of working.

(b) Briefly explain the working of an oscillator employing a UJT. Derive the expressions for the frequency of triggering firing angle delay in terms of η, and the charging resistance values.

Q.9 Write short notes on any two of the following: **[14]**

(a) Uninterrupted power supply

(b) Regenerative braking control

(c) Speed control of three-phase induction motor

(d) HVDC transmission

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